VISHWAS · SIH26079 · TEAM HELLFIRE LABS

Ankit

Ground truth and the ensemble concept

Final prep — script, judge Q&A, and every number traced

B.Tech | Slide 3 — you open it | about 35 seconds of speaking

YOUR JOB

You have one job and it is a good one: you make the rest of slide 3 make sense. Two ideas — what an ensemble actually is, and why we score our forecasts against real rain gauges instead of the easier option. The second of those is one of the strongest honesty arguments on the whole deck, and it is yours to land. Learn these two ideas properly rather than learning the whole slide badly. Devansh takes every implementation question — that is agreed in advance, and handing one over is the plan working, not you failing.

1 RUNNING ORDER

Mahika leads. Your rows are highlighted.

#	SPEAKER	WHAT HAPPENS	TIME
1	Mahika	Slide 1 — opens. Team, problem statement, the hook.	~40s
2	Arnav	Slide 2 — what VISHWAS is, and why the forecast underneath decays.	~70s
3	Devansh	Slide 2 — the measurement, and the result.	~60s
4	**Ankit**	**Slide 3 — opens it: what an ensemble is, and why we score against rain gauges.**	**~35s**
5	Devansh	Slide 3 — the pipeline, the stack, features, label, model, the map.	~75s
6	Mahika	Bridge into slide 4.	~10s
7	Paras	Slide 4 — feasibility, the numbers, the three defects.	~65s
8	Arnav	Slide 4 — viability, the four risks, the precision choice.	~40s
9	Mahika	Slide 5 — the three interviews.	~45s
10	Dhruv	Slide 5 — who it serves, MoES's numbers, the benefits. Then slide 6 — sources.	~75s
11	Devansh	Slide 6 — our own measurement.	~20s
12	Mahika	Closes. The tagline, then into questions.	~20s

2 THE STORY — EVERYONE MUST BE ABLE TO SAY THIS

If a judge stops you in the corridor. Thirty seconds. Learn the shape, not the words.

Every weather forecast India issues arrives with no honesty label.

Forecasters judge confidence today by ensemble spread — run the model 11 times from slightly different starting conditions, and if the runs disagree, distrust the forecast. We measured that method against India's own rain gauges, across all four seasons and three years.

Compared crudely it looks strong: ensemble spread appears to explain 43.9% of the variance in forecast error. Compared fairly — within the same rainfall conditions — it explains under 1%.

So we built the replacement. On that same fair comparison it scores 0.565 and VISHWAS scores 0.728, on a test year the model had never seen.

Most teams pitch a hypothesis. We pitch a measurement.

3 WHAT CHANGED IN YOUR SCRIPT

Your wording is your own. Only these points moved.

- Q&A OWNERSHIP — you now own the domain question outright: why land only, why only a quarter of India. Arnav and Devansh no longer carry a version of it, so nobody will answer over you. You also own why ensemble spread fails. Your reproducibility answer is gone; that goes to Devansh.
- CORRECTED — “the forecasting centre runs the same model 11 times” now reads “many times over... in the archive we use, that is 11 runs”. Operational GEFS carries 31 members. The panel runs these systems; an unscoped 11 is the kind of detail they correct from the floor.
- Arnav no longer explains what an ensemble is before you. That explanation is yours alone now, so take it at your own pace.
- Only your closing line changed, so it matches the exact cue Devansh is listening for.
- One optional clause added — that IMD is the ministry's own product. It lands well with a judge from that ministry, so use it if the pace allows.
- Nothing else. Yours was the most accurate script of the six — the ensemble explanation in particular is correct, and Arnav's has been corrected to match yours.

4 YOUR SCRIPT

Grey boxes are stage directions — do not say them. Lines tagged TRIM are safe to cut if you are running long.

SLIDE 3 — THE OPENING

▸ *Devansh hands to you at the end of Slide 2. You take about 35 seconds, then he takes the rest of this slide.*

> Before Devansh explains the system, two things we need to understand.

> First — what is an ensemble?

> The atmosphere is chaotic, so we can't know its exact initial state.

> That's why a forecasting centre runs the same model many times over, each from slightly different initial conditions. In the archive we use, that is 11 runs.

If they agree, confidence is high. If they disagree, uncertainty is high.

This disagreement is called ensemble spread — the key uncertainty signal our project evaluates.

Second — our ground truth.

We evaluate forecasts against IMD's gauge-gridded rainfall data — actual observations from rain gauges.

TRIM And it is IMD's own product — from the ministry that owns this problem statement.

We could have used ERA5 reanalysis, but it is model-based and may share physics with the forecast system.

So, to avoid making the forecast look better than it actually is, we chose actual gauge observations.

Gauges are the harder target — and they are the honest one.

▸ *Stop there. Those last words are Devansh's cue. Do not walk through the six pipeline boxes — that is his.*

5 EVERY NUMBER YOU SAY, AND WHERE IT COMES FROM

If a judge challenges a figure, give the source — not a defence.

CLAIM	SOURCE / WHAT IT REALLY MEANS
11 members	GEFS v12 reforecast carries 11 on the dates we use — a control run plus 10 perturbed.
IMD 0.25° gauge-gridded rainfall	India Meteorological Department, 1901–2024. The Ministry's own product, and a gauge observation rather than a model output.
'a reanalysis is itself a model product'	This is the argument, not a figure. ERA5 assimilates physics that overlaps GEFS, so it shares errors with the thing being judged.

6 QUESTIONS THE JUDGES WILL ASK YOU

Yours to answer. Read them out loud once — an answer you have said aloud comes back under pressure; one you have only read does not.

Q1 What is an ensemble, in one sentence?

Running the same weather model about 11 times from slightly different starting conditions, because the atmosphere is chaotic and you never know the starting state exactly.

How much those runs disagree is called ensemble spread, and it is what forecasters use today to decide whether to trust a forecast.

Q2 Why not use ERA5 reanalysis? It has full coverage and no gaps.

Because a reanalysis is a model product. It assimilates observations, but its physics partly overlaps the physics of the forecast we are judging — so it shares errors with the thing being measured, and scoring against it flatters the forecast.

Rain gauges do not share anything with the forecast. They are just what fell. It is the harder target, it costs us coverage, and it is the reason the number means something.

It is also the Ministry's own product, which matters when we are presenting a result about their forecasts.

⚠ This is your strongest answer. If you get any opening at all, this is the one to give.

Q3 **Isn't gauge data sparse and noisy?**

> It is sparser than a reanalysis, yes — and that is exactly the cost we chose to pay. It covers land only, and only where there are gauges, which is why our domain is 1,252 of 5,041 gridpoints.
>
> We put that limitation on slide 4 ourselves rather than let it be found. What we get in exchange is a truth source that is independent of the forecast.

Q4 **Why is your domain only a quarter of India? / Why land only?**

> 1,252 of 5,041 gridpoints — 24.8%. That is the gauge domain. The cut happens twice: 621 points fall outside IMD's own box on geometry, and the rest go to the land mask, because IMD's gridded rainfall is built from rain gauges and ocean and no-gauge cells arrive as minus 999.
>
> It is a consequence of the truth source, not a modelling shortcut. We could have scored against a reanalysis and covered the ocean, but a reanalysis shares model physics with the forecast being judged. We chose the harder, honest target.

⚠ This whole question is yours now — Arnav and Devansh no longer carry a version of it. If it turns into whether the ungauged part is a GUESS, that is a distribution check Devansh ran: hand it to him.

Q5 **What does the data cost?**

> Nothing. The forecast archive is NOAA's public bucket with anonymous access and no credentials. The IMD rainfall comes through a public package. No GPU, no paid API.
>
> That matters for whether an operational centre could actually adopt this.

Q6 **If ensemble spread is what forecasters use, why does it not work?**

> Because across the whole country it mostly measures geography rather than uncertainty. Wet regions have both large spread and large errors, so pooled together spread looks like it explains 43.9% of the variance in forecast error.
>
> Compare forecasts in the same rainfall conditions and it drops under 1%. Devansh can take you through how that was measured.

7 IF IT IS NOT YOURS, HAND IT OVER

Naming the right teammate reads as a team that knows its own work. Guessing reads as one that does not.

IF THEY ASK ABOUT…	SAY THE NAME
How the data is fetched — GRIB, S3, byte-range reads, index sidecars	Devansh
Why only Wednesdays, 50.3 GB, how long it took, resumability	Devansh
How the grids are colocated, the 0.5° choice, the −999 masking	Devansh
Features, the bust label, the model, SHAP, calibration	Devansh
Anything about the score, the margin or the ablations	Devansh
Metrics, the alert budget, the defects we found	Paras
	Arnav

IF THEY ASK ABOUT…	SAY THE NAME
Viability, the risks, why GEFS and not NEPS, one test year	
A real example of a forecast bust	Arnav
Reproducibility, and whether it runs on live forecasts	Devansh
Users, interviews, impact, sources	Mahika or Dhruv

SITUATION	WHAT TO SAY
You don't know the answer	“I don't have that number to hand and I'd rather not guess — can I come back to you on it?” That is a strength for this team, not a weakness: when we could not trace one of our own figures, we removed it rather than present it.
It's a teammate's area	“That's <name>'s part — <name>, do you want to take that?” Say the name. Don't answer for them.
A judge challenges a number	Give the source, not a defence. Every figure on the deck traces to a government source, a peer-reviewed paper, or our own logs.
A judge says 'you're just predicting heavy rain'	Anyone can answer this one: we removed the rainfall forecast from the model entirely and the score went UP, 0.728 to 0.731.

8 NEVER SAY THIS

Every one of these was a real error caught during the build, or a rule set deliberately.

- × Never say '43.9% of forecast error'. It is 43.9% of the VARIANCE in forecast error. This is the single easiest thing to get wrong under pressure.
- × Never quote the margin as '+0.163'. Say '+0.13 to +0.16'.
- × Never say 'deciles' or 'the wettest tenth' about the model result. It is EIGHT bins. (Deciles are correct only for the separate Phase 1 correlation analysis.)
- × Never quote any accuracy figure. Base rate is about 6%, so accuracy is meaningless — and quoting it signals a team that does not understand imbalance.
- × Never say 'AI-adjusted forecast' or imply we correct the forecast. We DETECT busts. We do not correct anything.
- × Never say the pooled ROC-AUC pair (0.909 vs 0.885) — not on a slide, not in a figure, not aloud, not even if a judge asks directly. Answer with the pooled MARGIN, +0.023, which is published on slide 4. Quoting the pair as performance would commit, in our own work, exactly the error our pitch accuses the incumbent of.
- × Never quote +0.663 as 'the baseline we beat', and never quote a margin range containing +0.035 (that one came from a filter conditioning on the outcome).
- × Never call the regions 'IMD official regions'. They are approximate lat/lon boxes.
- × Never present the confidence map's date as evidence of general performance. It is ONE date, illustrating the form of the product.
- × Never convert the three field interviews into percentages or 'N% of users said'. They are three exploratory conversations.
- × Never quote 98%→46%, ₹13,331 cr, '50% faster', '~30% improvement', or a 0.34–0.37 false-alarm ratio. All were in earlier drafts; none could be traced; all were removed. Do not say any of these numbers out loud even while explaining that you removed them.
- × Never say SHAP and permutation importance 'agree on the drivers'. They pick the same top three in a different order.

9 NUMBER CHEAT SHEET

Same in all six packs. If two of you quote a figure differently in the same room, that is the damage.

WHAT	NUMBER	WATCH OUT
Spread explains, pooled	43.9% of the VARIANCE in forecast error	not '43.9% of forecast error'
Spread explains, fair comparison	0.17%–0.90% → say 'under 1%'	never quote +0.041 alone
VISHWAS score (stratified ROC-AUC)	0.728	—
Ensemble spread score	0.565	—
Margin	+0.13 to +0.16 · 95% CI +0.108 to +0.193	never '+0.163' alone
Matched 10% alert budget	VISHWAS catches 64% of busts vs 55%	—
POD / FAR / CSI / ETS at 10%	0.641 / 0.586 / 0.336 / 0.300 vs 0.546 / 0.647 / 0.273 / 0.235	—
Features	23, all from GEFS, none from the truth	not 24
Ensemble members	11 (control + 10 perturbed)	—

WHAT	NUMBER	WATCH OUT
Domain	1,252 gauge points of a 5,041-point analysis box (24.8%)	the 5,041 box is land AND ocean; only the 1,252 is land
Data fetched	50.3 GB · 104 forecast dates (2018-2019)	this study only; both studies together are 124.5 GB over 256 dates
Labelled rows	1,302,080 in this study · 651,040 in the test set	don't conflate the two; across both studies it is 3,205,120
Five-year retest	margin +0.149, 95% CI +0.116 to +0.203, 104 forecast dates	quote the MARGIN only, never an AUC beside 0.728
Live cycle scored	GEFS 20260907 and 20260908, ~100 s, ~210 MB, CPU	the live run is DONE — never call it future work
High-risk share	7.78% gauge-verified · 14.06% unverified part of the shipped map	ALWAYS name the domain; never blend them
Train / test	trained on 2018 · tested on all 52 Wednesdays of 2019	—
Base rate	about 6%	so never quote accuracy
Bust definition	|forecast − observed| above the 95th percentile, per region and lead time, threshold set on training data only	—
Rainfall bins	EIGHT bins · 40.92% bust rate in the wettest, 0.15% in the driest	never 'deciles' or 'wettest tenth'
Ablations	full 0.728 · no month 0.722 · no rainfall forecast 0.731 · no 850 shear 0.726 · spread-only 0.562	—
Underlying GEFS, heavy rain (≥64.5 mm)	detects 23% of events at Day 1, 0.2% at Day 10	this is the forecast, NOT our model
IMD warning accuracy (MoES)	85, 73, 67, 63, 58% for Day 1–5	Lok Sabha Q.2953, 17 Dec 2025
The tie	ensemble spread is exactly 0.00 mm on 189,758 test rows — a third of the set	—
Calibration	Brier 0.04217 vs 0.04749 · resolution 1.4× the incumbent's	—

ONE LAST THING

The strongest thing about this project is not the score. It is that we measured the method used today before proposing a replacement, and that we removed our own figures when we could not source them. If you land one impression, land that one.